

Document Title:	Heracles Internal and External GUI SFMEA			
Document Number:	DHF-HRC-0028	Revision:	1	
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Heracles Internal and External GUI Software Failure Modes and Effects Analysis

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1: Purpose

The purpose of this Software Modes and Effects Analysis (SFMEA) is to evaluate the potential failure modes in the **Internal and External GUI Software**, understand their effects on the system, patient, and other users, prioritize risks, and recommend corrective actions to mitigate those risks. The purpose is to ensure the safety and effectiveness of the device throughout its lifecycle.

2: Scope

The scope of this SFMEA covers the Internal and External GUI Software, derived from its design. This document to be continuously maintained and updated throughout the lifecycle of the medical device, as potential software failure modes are addressed and mitigated to achieve an acceptable or As Low As Possible level of risk.

3: Abbreviations and Definitions

Abbreviation/Term	Definition
ALAP	As Low As Possible
SDD	Software Design Document
SFMEA	Software Failure Modes and Effects Analysis
SW	Software

4: Overview

The Internal GUI console provides a means for the user to maintain demographic patient data, including treatment plans, and perform imaging. It communicates with the record and verify component, flange camera stream, imaging component and robotic arm controller.

The internal GUI software runs on the All-In-One console located on the base of the system.

The external GUI console provides a means for the user to activate and monitor the treatment. It communicates with the main control board, treatment head camera and imaging head camera.

The external GUI software runs on the external tablet console located outside the room.

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5: References

Document ID / Reference Number	Document Title
DHF-HRC-0033, Rev. 1	Heracles Internal GUI Software Design Document (SDD)
DHF-HRC-0042, Rev. 1	Heracles External GUI Software Design Document (SDD)
QSP-017, Rev. 1	Risk Management Procedure

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6: SFMEA Matrix

Design Item ID	Design Item Title	Failure Mode	Foreseeable Sequence of Events	Effect	Causes	Initial Probability of Occurrence – P	Initial Severity of Harm - S	Initial Risk Index (RI) / Initial Risk Acceptability	Risk Controls Description	Residual Probability of Occurrence - P	Residual Severity of Harm - S	Residual Risk Index (RI) / Residual Risk Acceptability
H10SG-683	Calculate Emission Duration	Miscalculation of Emission Duration	The calculation logic has a bug	Over-exposure to radiation	Defect in software	3 - Probable	4	12/ Zone 2 ALAP	1. Firmware will reject the treatment plan if the duration is equal or more than 300 sec 2. Unit test + code review	1 - Improbable	4	4 / Zone 1 Acceptable
H10SG-683	Calculate Emission Duration	Miscalculation of Emission Duration	The calculation logic has a bug	under-exposure to radiation	Defect in software	3 - Probable	2	6 / Zone 1 Acceptable	Unit test + code review	1 - Improbable	2	2/ Zone 1 Acceptable
H10TG-276	Validate Active Collimator	Failure to validate active collimator	Failure in collimator identification cause the system to treat patient with wrong collimator	Over-exposure to radiation	Defect in software	3 - Probable	4	12/ Zone 2 ALAP	Unit test + code review	1 - Improbable	4	4/ Zone 1 Acceptable
H10TG-276	Validate Active Collimator	Failure to validate active collimator	Failure in collimator identification cause the system to treat patient with wrong collimator	Under-exposure to radiation	Defect in software	3 - probable	2	6/ Zone 1 Acceptable	Unit test + code review	1 - Improbable	2	2/ Zone 1 Acceptable
H10TG-263	Stop Emission	Stop Emission Failed	User pressed the stop button and emission fails to stop	Over-exposure to radiation	Defect in software	3 - probable	4	12/ Zone 2 ALAP	1. E-Stop physical switch 2. Unit test + code review	1 - Improbable	4	4/ Zone 1 Acceptable

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H10TG-257	Record Treatment Delivered Dose	Failure to Record Delivered Dose after treatment	Delivered Treatment Dose is not recorded after treatment Treatment data does not exist in the next session	no data recorded, impacting future treatments that might result in Over-exposure to radiation	Database communication failure	3 - probable	4	12/ Zone 2 ALAP	1. Data will be saved to file and can be restored 2. Alert prompt of data not saved is displayed to the user 3. Unit test + code review	1 - Improbable	4	4/ Zone 1 Acceptable
H10TG-257	Record Treatment Delivered Dose	Failure to Record Delivered Dose after treatment	Delivered Treatment Dose is not recorded after treatment Treatment data does not exist in the next session	no data recorded, impacting future treatments that might result in Over-exposure to radiation	Defect in software	3 - probable	4	12/ Zone 2 ALAP	Unit test + code review	1 - Improbable	4	4/ Zone 1 Acceptable
H10TG-257	Record Treatment Delivered Dose	Record Faulty Data	Incorrectly delivered dose recorded in the database	no data recorded, impacting future treatments that might result in Over-exposure to radiation	Defect in software	3 - probable	4	12/ Zone 2 ALAP	Unit test + code review	1 - Improbable	4	4/ Zone 1 Acceptable
H10TG-255	Beam On process	Failure to start emission	User pressed the Beam-On button and the emission fails to start	Treatment is not delivered	Hardware-software communication failure	3 - probable	1	3 / Zone 1 Acceptable	Unit test + code review	1 - Improbable	1	1/ Zone 1 Acceptable

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H10TG-274	Retrieve treated patient details	Wrong patient data sent to the application	The operator loads patient data to the external console and the external console displays wrong patient data	Wrong patient details and treatment plan are displayed that might result in over-exposure to radiation	Defect in software	3 - probable	4	12 / Zone 2 ALAP	Unit test + code review	1 - Improbable	4	4/ Zone 1 Acceptable
H10SG-683	Calculate Emission Duration	Miscalculation of Emission Duration	The calculation logic has a bug	Over-exposure to radiation	Defect in software	3 - probable	4	12 / Zone 2 ALAP	1. Firmware will reject the treatment plan if the duration is equal or more that 300 sec 2. Unit test + code review	1 - Improbable	4	4/ Zone 1 Acceptable
H10SG-683	Calculate Emission Duration	Miscalculation of Emission Duration	The calculation logic has a bug	under-exposure to radiation	Defect in software	3 - probable	2	6 / Zone 1 Acceptable	Unit test + code review	1 - Improbable	2	2/ Zone 1 Acceptable
H10TG-276	Validate Active Collimator	Failure to validate active collimator	Failure in collimator identification cause the system to treat patient with wrong collimator	Over-exposure to radiation	Defect in software	3 - probable	4	12 / Zone 2 ALAP	Unit test + code review	1 - Improbable	4	4/ Zone 1 Acceptable

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7: SFMEA Risk Prioritization

The Risk Index (RI) is a critical factor in prioritizing corrective actions. It provides a relative indication of the identified failure indicating whether the risk requires further mitigation or control.

RI is calculated as follows (see QSP-017 risk management, referenced in section 5):

- **RI = Probability of Occurrence (P) X Severity (S)**

Using $P \times S = RI$		Severity Rank				
		Negligible	Minor	Major	Critical	Catastrophic
Occurrence Rank		1	2	3	4	5
Highly Likely	5	Zone 2	Zone 2	Zone 3	Zone 3	Zone 3
Very Likely	4	Zone 1	Zone 2	Zone 2	Zone 3	Zone 3
Probable	3	Zone 1	Zone 1	Zone 2	Zone 2	Zone 3
Rarely	2	Zone 1	Zone 1	Zone 1	Zone 2	Zone 2
Improbable	1	Zone 1	Zone 1	Zone 1	Zone 1	Zone 2

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8: SFMEA Risk Acceptability

The following Risk Zones define the 3 levels of Risk Acceptability:

Risk Zone	Risk Evaluation Actions Method
Unacceptable Zone 3	Reduce risks within this region to a Zone 2 or Zone 1 risk with control measures or design improvements. If none of these are feasible, document that the risk has been reduced as far as possible and evaluate whether the benefit outweighs the residual risk or eliminate the feature or design completely.
ALAP Zone 2	Reduce all Zone 2 risks with risk control measure(s). If the Zone 2 risk directly results from design and/or manufacturing failures, pursue additional controls, or design improvements to reduce the risk. If none of these are feasible, document that the risk has been reduced as low as possible and evaluate whether the benefit outweighs the residual risk.
Acceptable Zone 1	Within this region, the level of risk is considered low and considered to be acceptable once the risk has been reduced as far as possible. Reduction of risk to as far as possible can be demonstrated through conformance to requirements or specification and/or implementation of risk control measure in the risk management documents.

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9: Record History

Revision	Rev Date	Author	Description of Changes
1	05-MAR-2025	Limor Erez	Initial Revision

10: Record Review and Approval

Author	Author Function	Signature and Date
Limor Erez	Director, Software & Products	 Electronically signed by: Limor Erez Reason: I am the author Date: Mar 5, 2025 22:15 GMT+2

Reviewer/ Approver(s)	Function	Signature and Date
Yonatan Vainer	CTO / PMO	 Electronically signed by: Yonatan Vainer Reason: I am the approver Date: Mar 5, 2025 22:23 GMT+2
Adi Fishman	Principal Software Quality Systems Engineer / Quality Assurance	 Electronically signed by: Adi Fishman Reason: I am the approver Date: Mar 5, 2025 15:52 EST

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